

CSCI-566 Deep Learning Project

Hallucination-Resistant Medical Question Answering with Patient-Specific Memory

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Abstract

Large Language Models (LLMs) achieve strong performance in question answering but remain prone to hallucinations, producing fluent yet unsupported or incorrect responses. This issue is particularly critical in the medical domain, where inaccurate patient-specific information can compromise safety. In this project, we frame hallucination as a system-level design problem rather than solely a model limitation; by "system-level" we mean constraints enforced programmatically outside the model's forward pass, including retrieval filtering, evidence thresholding, and abstention logic. We propose a hallucination-resistant framework that combines Retrieval-Augmented Generation (RAG) with explicit evidence constraints and an abstention mechanism over structured patient records. The system is designed to generate answers only when sufficient supporting evidence is available; otherwise, it abstains to reduce unsupported outputs. We conduct a pilot evaluation against multiple baselines using hallucination-focused metrics such as hallucination rate, evidence grounding, and supported answer rate. Current results are based on Gemini and GPT-5.4 outputs.

1 Introduction

In this project, we investigate whether explicit evidence constraints can eliminate unsupported claims in medical question answering without making responses unusably incomplete. We propose a hallucination-resistant framework that combines retrieval augmented generation (RAG) with programmatic evidence constraints and an abstention mechanism. The system operates over structured patient records and generates answers only when sufficient supporting evidence is retrieved; otherwise, it abstains with responses such as "Insufficient information." This shifts responsibility for correctness from the model's parametric knowledge to an external evidence verification pipeline grounded in patient data. We compare our approach against full-context prompting, rule-based retrieval, and standard semantic RAG, evaluating across both Gemini and GPT models. Our contributions include a modular evidence-constrained pipeline, a hallucination-focused evaluation setup, and initial results demonstrating reduced unsupported content. The remainder of this report covers our current research progress (Section 2), literature review (Section 3), experimental results (Section 4), and updated research directions (Section 5).

2 Current Research Progress

We have implemented a modular patient-centric QA pipeline and partially evaluated it for hallucination detection and evidence-grounded answering. The system separates retrieval, prompt construction, model invocation, and evaluation so that ground-

ing and assessment are handled explicitly outside the language model. Initial end-to-end pilot experiments have been run across multiple LLM backends to validate the pipeline and measure qualitative behavior of proposed defenses against hallucination.

2.1 Dataset

The experimental corpus is a de-identified structured medical QA dataset containing 200 patient records. The dataset was synthetically generated by the authors, with patient records and QA interactions constructed to reflect realistic clinical scenarios across a range of question types. Each record includes structured clinical fields and associated QA items with evidence annotations. Each patient record contains standard clinical fields (demographics, past medical history, active medications, documented allergies, and current symptoms) together with a set of gold QA interactions.

2.1.1 Gold QA Format

For each QA item the dataset provides: (i) a question, (ii) a ground-truth answer, and (iii) fine-grained supporting evidence entries that explicitly point to the source clinical field and the exact evidence text. This provenance annotation enables automated, evidence-aware evaluation of model outputs (measuring whether a claim is supported by the patient record).

2.1.2 Properties and Suitability

Records are strictly patient-local (no mixing of information across patients) and therefore appropriate for medically sensitive experiments. The dataset is balanced across a variety of clinical question types (presence/absence queries, medication checks, allergy contraindications, simple factual retrieval), which permits measuring both hallucination (unsupported claims) and standard QA metrics (exact match, token F1). Because each QA item links answers to explicit evidence spans, the data supports precision/recall style metrics for grounding and can be used for abstention-aware evaluations.

2.2 Completed Tasks

2.2.1 Modular system architecture

We implemented a modular pipeline that enforces separation of concerns: (1) patient data loading and chunking, (2) retrieval (rule-based and semantic), (3) baseline prompt construction, (4) a unified model invocation layer that can route prompts to different LLM backends, and (5) automated evaluation producing per-query metrics and aggregate summaries.

2.2.2 Retrieval components

We implemented two retrieval strategies: a rule-based keyword retriever as an interpretable baseline, and a semantic retriever using all-MiniLM-L6-v2 (384-dimensional sentence embeddings) with cosine similarity over atomically chunked patient fields,

retrieving the top $k = 5$ most relevant chunks per query.

2.2.3 Baselines and proposed method

Three baselines were implemented: (1) full-context prompt containing all patient fields, (2) rule-based retrieval prompt with filtered fields, and (3) semantic RAG using top- k chunks. The proposed method augments semantic RAG with an explicit prompt-level constraint instructing the model to answer only if the retrieved evidence explicitly supports a response; otherwise the model returns 'Insufficient information', which is detected programmatically via keyword matching on a fixed set of abstention phrases. This enforces evidence grounding at the system level.

2.2.4 Model integration and pilot runs

A unified query function was used to run pilot experiments across hosted and local LLMs (commercial API and Hugging Face models). Pilot runs on held-out subsets validated end-to-end behavior, response formats, and metric computation.

2.2.5 Evaluation suite

A set of objective metrics was developed and applied per query: hallucination rate (fraction of model claims unsupported by any gold evidence), exact match and token-level F1 against gold answers, evidence precision/recall/F1 (over tokens or spans), supporting answer rate (SAR), and abstention rate. Aggregation and visualization utilities were created for cross-model and cross-baseline comparisons.

2.3 Ongoing Work

2.3.1 Performance bottlenecks and mitigation

Large-scale experiments are currently constrained by API rate limits and per-call latency on hosted LLMs; this increases wall-clock time and cost for full-cohort evaluation. To address this we are implementing (a) intelligent request batching and concurrency control with adaptive rate limiting and exponential backoff, (b) deterministic response caching to avoid repeated identical calls during iterative development, and (c) a hybrid experiment strategy that uses local open models for broad sweeps and hosted APIs for a focused subset of critical comparisons.

2.3.2 Experimental refinements

We are finalizing ablation experiments to quantify sensitivity to retrieval hyperparameters (chunk length, chunk overlap, embedding model, top- k), prompt template variations (instruction phrasing and constraint wording), and the evidence threshold that triggers abstention. Each ablation will report 95% confidence intervals computed via bootstrap resampling ($n = 1000$) over per-query metric scores; this is planned as an extension to the current aggregation pipeline.

2.3.3 Validation and human review

Automated metrics will be supplemented by targeted human annotation for ambiguous or high-impact cases to calibrate automatic hallucination detectors and refine evaluation thresholds. We will also sample model outputs for qualitative error analysis (types of hallucination, partial grounding, hallucinated contraindications).

2.3.4 Planned deliverables and timeline (2–4 weeks)

(1) stabilize rate-limit mitigations and run full-cohort experiments using the hybrid strategy, (2) complete ablation studies

and compute aggregate metrics with uncertainty estimates, (3) produce figures and tables comparing baselines and models (hallucination rate, evidence precision/recall, SAR, abstention trade-offs), and (4) draft a results section with quantitative evidence that the proposed evidence-constrained RAG reduces unsupported claims at the measured cost of increased abstentions.

3 Literature Review

The literature relevant to our project falls into four themes: domain-specialized medical LLMs, retrieval-grounded medical QA, hallucination detection and abstention, and faithful reasoning under evidence constraints.

[1] The paper "Clinical Camel: An Open Expert-Level Medical Language Model with Dialogue-Based Knowledge Encoding" presents an open medical LLM built by fine-tuning LLaMA-2 with QLoRA and dialogue-based knowledge encoding. It shows that domain-specific fine-tuning can substantially improve medical question answering across benchmarks such as PubMedQA, MedQA, and MedMCQA. This work is important because it demonstrates the value of adapting general LLMs to the medical domain, although it focuses more on capability than hallucination control.

[2] "Toward Expert-Level Medical Question Answering with Large Language Models" introduces Med-PaLM 2, a medically specialized model built on PaLM-2 with domain-specific tuning and refinement. The paper reports strong benchmark performance and physician-level answer quality in human evaluations, highlighting the promise of large medical LLMs. However, it also suggests that high capability alone is not sufficient for safe real-world deployment without additional grounding mechanisms.

[3] "Smoothing Out Hallucinations: Mitigating Hallucinations in LLMs via Softmax Smoothing" studies hallucination reduction through smoothed knowledge distillation. By reducing model overconfidence, the method improves factual consistency while preserving general language understanding performance. This paper is useful because it treats hallucination mitigation as a training and calibration problem rather than only a retrieval problem.

[4] "Towards Reliable Medical Question Answering: Techniques and Challenges in Mitigating Hallucinations in Language Models" directly addresses hallucination in medical QA. It argues that retrieval-augmented generation, explicit evidence grounding, and abstention when evidence is insufficient are central to building reliable systems. This work is especially close to our project because it frames hallucination as a safety issue and motivates evidence-constrained answering.

[5] "Benchmarking Retrieval-Augmented Generation for Medicine (MIRAGE)" provides a systematic comparison of RAG methods for medical QA. It shows that retrieval can improve answer accuracy significantly, but also notes that retrieving too much information can introduce noise and hurt performance. This paper is highly relevant because it supports the idea that retrieval quality and context selection matter as much as the model itself.

[6] "MedTrust-RAG: Evidence Verification and Trust Alignment for Biomedical Question Answering" proposes a citation-grounded framework in which answers must be supported by

evidence and the system refuses to answer when support is insufficient. Its two-agent design separates evidence verification from answer generation, improving trustworthiness. This paper closely aligns with our project’s focus on grounded generation and abstention.

[7] “MEGA-RAG: A Retrieval-Augmented Generation Framework with Multi-Evidence Guided Answer Refinement for Mitigating Hallucinations of LLMs in Public Health” extends standard RAG by combining dense, sparse, and knowledge-graph retrieval, followed by candidate comparison and refinement. The main contribution is stronger evidence grounding through multi-source verification. This motivates richer retrieval and verification strategies for reducing unsupported answers.

[8] “DKITNLP at ArchEHR-QA 2025: A Retrieval Augmented LLM Pipeline for Evidence-Based Patient Question Answering” focuses on patient-specific QA over clinical notes. It retrieves minimal sentence-level evidence using BioBERT retrieval and reranking, then generates grounded answers with an instruction-tuned LLM. This is very relevant to our work because it shows how fine-grained retrieval over patient records can improve traceability and reduce hallucination.

[9] “MedInsight: A Multi-Source Context Augmentation Framework for Generating Patient-Centric Medical Responses Using Large Language Models” combines patient context with external knowledge to generate personalized medical responses. Its emphasis is more on patient-friendly explanation and multi-source context fusion than on strict evidence verification. This makes it useful as a comparison point for systems that prioritize helpfulness over strong safety constraints.

[10] “Healthcare Agent: Eliciting the Power of Large Language Models for Medical Consultation” studies agentic medical consultation systems with memory and multi-turn reasoning. It shows that conversation memory and workflow design can improve follow-up quality, continuity, and safety. This paper is relevant because it highlights that reliability can come not only from model training but also from system design and interaction structure.

[11] “HalluLens: LLM Hallucination Benchmark” proposes a benchmark that evaluates not just correctness but also abstention behavior. It argues that a reliable model should know when to refuse to answer, and therefore hallucination evaluation must include false refusals and false acceptances. This directly supports our project’s abstention-aware evaluation.

[12] “LINS: A General Medical Q&A Framework for Enhancing the Quality and Credibility of LLM-Generated Responses” argues that retrieval alone is insufficient unless evidence is filtered, decomposed, and explicitly linked to statements. By using statement-citation alignment and multi-agent decomposition, it improves answer credibility. This paper supports our emphasis on explicit evidence tracing rather than simple context stuffing.

[13] “TruthfulQA: Measuring How Models Mimic Human Falsehoods” is a broader benchmark on model truthfulness. It shows that larger models may still reproduce common misconceptions and that truthfulness must be measured separately from fluency. Although not medical-specific, it is important because it motivates why factuality and safety need dedicated evaluation metrics.

[14] “SelfCheckGPT: Zero-Resource Black-Box Halluci-

nation Detection” proposes detecting hallucinations using self-consistency across multiple sampled outputs. If a model gives inconsistent answers to the same question, the answer is more likely to be unreliable. This is useful for our project as a possible lightweight post-generation hallucination detector.

[15] “Semantic Entropy for Hallucination Detection in LLMs” introduces semantic entropy as a way to measure uncertainty at the meaning level rather than the token level. It shows that higher semantic uncertainty correlates with hallucination risk. This paper is relevant because it offers a principled uncertainty signal that could complement evidence-based abstention.

[16] “Localizing Before Answering: A Hallucination Evaluation Benchmark for Grounded Medical Multimodal LLMs” studies hallucination in multimodal medical QA. It proposes that models should first localize relevant medical regions before answering, which improves both grounding and robustness. This broadens the literature review by showing that hallucination is also a grounding problem in multimodal settings.

[17] “Reducing Hallucinations of Medical Multimodal Large Language Models with Visual Retrieval-Augmented Generation” reduces hallucination by retrieving similar medical images and associated reports. By grounding outputs in both visual and textual evidence, it improves reliability in multimodal medical tasks. This paper is useful because it extends the RAG idea beyond text-only pipelines.

[18] “MEDSEARCH: A Conversational Agent for Real-Time, Evidence-Based Medical Question-Answering” presents a multi-LLM medical QA system with real-time evidence retrieval. It uses task-specific routing and answer selection to improve consistency and maintain evidence-based reasoning. This paper is relevant because it shows how retrieval, orchestration, and verification can work together in practical medical QA systems.

[19] “MedRAG: Enhancing Retrieval-Augmented Generation with Knowledge Graph-Elicited Reasoning for Healthcare Copilot” incorporates diagnostic knowledge graphs into RAG to support more precise and evidence-grounded reasoning. It also supports proactive follow-up questioning when information is incomplete. This paper is important because it suggests that structured medical knowledge can improve both grounding and diagnostic support.

[20] “Beyond Correctness: Rewarding Faithful Reasoning in Retrieval-Augmented Generation” argues that correct final answers are not enough if the intermediate reasoning is unsupported. It introduces the idea of reasoning faithfulness, requiring consistency between retrieved evidence, reasoning steps, and final output. This is relevant to our project because a medical QA system should be both correct and evidence-faithful.

[21] “R-Tuning: Instructing Large Language Models to Say ‘I Don’t Know’” explicitly trains models to refuse answering when knowledge is insufficient. The paper shows that refusal behavior can generalize across domains and act as a transferable safety skill. This strongly supports our project’s abstention mechanism, where refusing to guess is often safer than generating unsupported medical advice. Overall, prior work shows that medical QA reliability depends not only on model capability but also on retrieval quality, evidence verification, abstention behavior, and uncertainty estimation. Our project is most closely related to work on evidence-grounded QA with abstention, but differs in focusing specifically on patient-record-level support.

4 Initial Results

We conducted a pilot end-to-end evaluation of the proposed hallucination-resistant pipeline against three baselines using two held-out patient records from the structured medical QA dataset (24 model-baseline evaluations total). The pilot aimed to validate the full system pipeline rather than yield statistically conclusive results. Initially, due to API rate limits (Gemini free tier), only the Gemini backend produced valid outputs. Subsequently, we also incorporated GPT-based evaluation, enabling comparison across both Gemini and GPT models. All referenced figures are auto-generated by the visualization module.

4.1 Hallucination Rate

Hallucination rate measures the fraction of responses containing content unsupported by the ground-truth answer or evidence passages. Across baselines, both models exhibited low hallucination rates, with Gemini consistently producing 0.00 across all baselines, while GPT showed a non-zero rate only for the proposed method (0.14). This reflects that while GPT occasionally generates unsupported content under stricter constraints, Gemini tends to be more conservative. These preliminary results suggest that explicit grounding constraints can reduce unsupported content. Figure 1 illustrates this comparison.

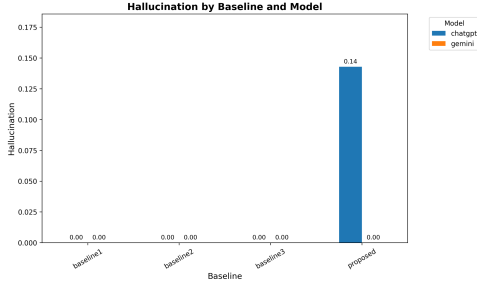


Figure 1: Hallucination Rate Comparison

4.2 Answer Quality (Token F1)

On token-level F1, both models performed better on baselines than the proposed method. GPT achieved higher F1 scores across all baselines (Baseline 1: 0.65; Baseline 2: 0.63; Baseline 3: 0.63) compared to Gemini (0.50, 0.50, and 0.58, respectively). However, for the proposed method, GPT scored 0.44 while Gemini scored 0.14. This gap reflects both model differences and a deliberate design trade-off. The proposed method produces terse responses (e.g., "Yes"), whereas ground-truth answers are full clinical sentences. Token F1 penalizes brevity heavily, even when the answer is factually correct. This motivates incorporating semantically aware metrics such as BERTScore in future evaluations. Figure 2 shows the F1 comparison.

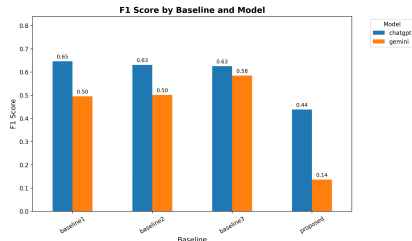


Figure 2: F1 Score Comparison

4.3 Evidence Grounding (Evidence F1)

Evidence F1 measures how well predicted answers are grounded in retrieved evidence, computed as the harmonic mean of evidence precision and recall. GPT outperformed Gemini across all baselines (approximately 0.50-0.51 vs. 0.23-0.39), indicating stronger alignment with retrieved evidence. For the proposed method, GPT achieved 0.33 while Gemini scored 0.00. As before, this is influenced by response brevity: short answers like "Yes" contain no content words above the threshold used for evidence recall, causing scores to drop regardless of factual correctness. This highlights a limitation of token-overlap evidence metrics rather than a failure of grounding, which is enforced structurally through prompt constraints. Figure 3 shows this comparison.

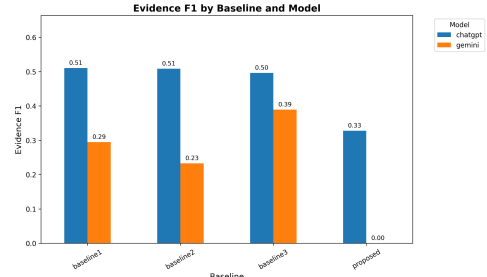


Figure 3: Evidence F1 Score Comparison

4.4 Exact Match, SAR, and Abstention

Exact match scores remained low across all configurations, which is expected for open-ended generative QA where outputs are paraphrases or shortened forms of ground-truth sentences. Gemini achieved a high exact match score for the proposed method (0.80), while GPT produced near-zero exact matches across settings, reflecting stylistic differences in response generation. For Supporting Answer Rate (SAR), GPT achieved higher values across baselines (approximately 0.57, 0.57, and 0.43) compared to Gemini (0.17, 0.17, and 0.33), while both models showed reduced SAR for the proposed method (GPT: 0.14, Gemini: 0.00). This indicates that stricter grounding reduces verbosity and supporting content in responses. Abstention behavior was minimal across baselines, with GPT showing a small abstention rate (0.14) for the proposed method while Gemini remained at 0.00 across all settings. This suggests that GPT is more likely to abstain under uncertainty, whereas Gemini tends to provide an answer regardless. These metrics are expected to become more discriminative in the full-cohort evaluation with unanswerable or ambiguous questions. Figures 4, 5 and 6 capture these distributions.

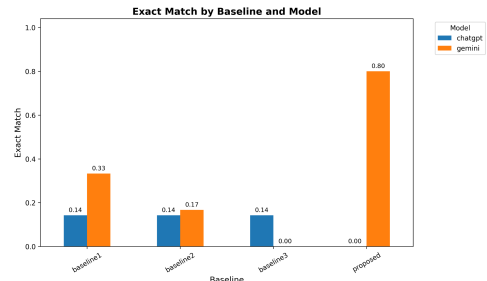


Figure 4: Answer Exact Match

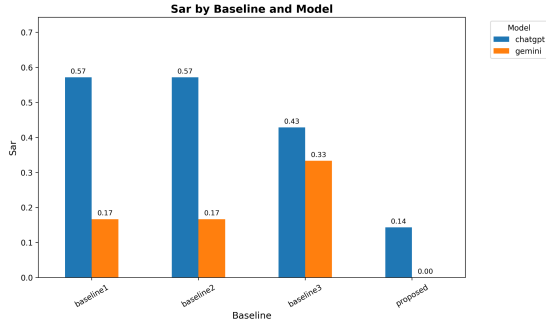


Figure 5: SAR

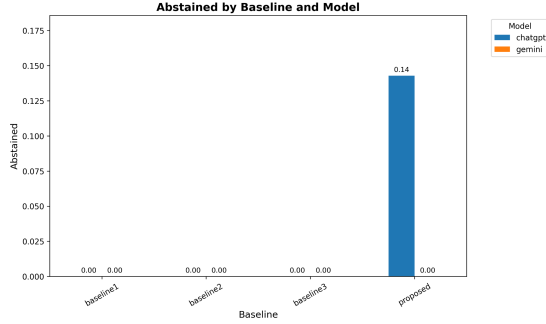


Figure 6: Abstained Score

4.5 Qualitative Analysis

For factual presence-or-absence questions (e.g., whether a medication is documented), all methods produced correct answers, with baselines generating verbose justifications and the proposed method responding concisely. A more revealing case arose for patient PT-46928690: "Is ibuprofen safe to prescribe for this patient?" The ground-truth answer cites a documented ibuprofen allergy causing gastrointestinal bleeding. All three baselines were flagged as hallucinating, not for giving the wrong yes/no answer, but for introducing additional clinical reasoning (e.g., NSAID platelet inhibition risks) that, while medically plausible, was unsupported by the patient record. This exemplifies the hallucination category our system targets: fluent, clinically reasonable content that lacks evidentiary support. The proposed method stayed within the bounds of retrieved evidence and was not flagged. This case illustrates why hallucination rate rather than token F1 is the most meaningful metric for safety-critical medical Q&A and why a method with lower F1 but zero unsupported content may be preferable in clinical settings.

4.6 Summary

The pilot provides preliminary evidence that evidence-constrained RAG can reduce unsupported content in this small evaluated subset. The primary cost is reduced response verbosity, which lowers token-overlap metric scores. Producing answers that are both concise and well-grounded while performing well on standard NLP metrics remains an open challenge that will guide our ablation studies and metric selection in the full evaluation.

5 Updated Research Ideas

Our current system retrieves patient-specific information and gives it to an LLM to answer medical questions. Since the

SUMMARY STATISTICS						
By Baseline:						
	hallucination	exact_match	f1_score	evidence_f1	sar	abstained
baseline						
baseline1	0.0000	0.2388	0.5766	0.4107	0.3846	0.0000
baseline2	0.0000	0.1538	0.5708	0.3811	0.3846	0.0000
baseline3	0.0000	0.0769	0.6065	0.4466	0.3846	0.0000
proposed	0.0833	0.3333	0.3129	0.1912	0.0833	0.0833
By Model:						
model	hallucination	exact_match	f1_score	evidence_f1	sar	abstained
chatgpt	0.0357	0.1071	0.5852	0.4606	0.4286	0.0357
gemini	0.0000	0.3043	0.4421	0.2391	0.1739	0.0000

Figure 7: Summary Statistics

main goal of the project is to reduce hallucinations, our next research ideas focus on making the system more reliable, better grounded in evidence, and safer in cases where the patient record is incomplete or unclear.

- Better Retrieval Methods:** We want to test different ways of retrieving patient information, such as giving the full patient record, using keyword-based retrieval, or using vector-based retrieval. This will help us understand which method gives the most relevant context and reduces hallucinations the most.
- Evidence-Based Answering:** Another idea is to make the model answer only when the retrieved patient data clearly supports the response. This can help reduce answers that sound correct but are not actually grounded in the record.
- Abstention When Evidence Is Missing:** In medical settings, it may be safer for the system to say "I don't know" instead of guessing. We plan to explore ways for the model to abstain when the patient data does not contain enough information to answer confidently.
- Lightweight Hallucination Detection:** To include a training-based component, we can build a small classifier that checks whether a generated answer is supported by the patient data or is hallucinated. This would allow us to add training without changing the overall system too much.
- Verification Layer:** We also want to explore adding a second step that verifies whether the final answer matches the retrieved evidence before it is shown to the user. This could work as an extra safety check.
- Comparing System Design vs Training:** Finally, we want to compare system-based improvements such as retrieval and prompting with lightweight training-based methods such as a classifier or verifier. This will help us understand whether better system design alone is enough, or whether some training is also needed to reduce hallucinations effectively.

Overall, our project is still centered on safe medical question answering, not diagnosis prediction. The main research direction is to study how retrieval, grounding, abstention, and lightweight training can work together to make the system more trustworthy.

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